



## Pune AI Community

4,182 followers

now • 

...

Gemma 4 is here 🚀 and the most interesting part is the new terminology around “effective” models.

- “Effective” means smarter parameter usage using per layer embeddings (PLE)
- The E2B model has 2.3 billion effective parameters, 5.1 billion total parameters.
- The PLE technique gives each decoder layer its own small embedding for every token.
- More intelligence per parameter without just scaling model size 🧠
- Small models now capable of strong reasoning and agentic workflows
- Designed to run locally on consumer GPUs and even edge devices
- E2B and E4B target mobile and offline multimodal use cases 📱
- 128K context for edge models, 256K for larger ones
- Apache 2.0 license and available across HuggingFace, LM Studio, Ollama, and more

Big takeaway: Extended “effective” models push serious AI capability into small, local, efficient deployments.

Edge AI just got more practical. ⚡

We are trying it locally and will keep you posted!!

[#AI](#) [#Gemma](#) [#OpenSourceAI](#) [#LocalLLM](#) [#EdgeAI](#) [#AgenticAI](#) [#GenerativeAI](#) [#AIEngineering](#) [Google](#) [Rajat Bhatia](#) [Google Developer Experts](#) [Google Developers Group](#) [GDG Pune](#) [GDG Cloud Pune](#)

# Gemma 4: The New **"Effective"** Models

The diagram illustrates the Gemma 4 model architecture. At the top, four colored cubes represent different model sizes: 2B (white), 4B (red), 26B MoE (orange), and 31B Dense (blue). Below these, a flowchart shows the model's components. On the left, a cloud icon is labeled 'Open Weights, Open Source'. Below it, a server and mobile devices are shown with the text 'Runs Locally on GPU & Edge Devices' and 'Mobile & Offline Multimodal'. In the center, a brain icon is labeled 'Smarter Parameter Usage' and 'Per Layer Embeddings'. To the right, a bar chart shows 'ELO Score' and a context window of '128K / 256K Context' with '128K' and '256K' labels. At the bottom, the text 'Available Now' is followed by logos for 'Apache 2.0 License', 'LM Studio', and 'Ollama'.

Open Weights,  
Open Source

2B 4B 26B MoE 31B Dense

Smarter Parameter Usage

Per Layer Embeddings

Runs Locally on GPU &  
Edge Devices

Mobile & Offline  
Multimodal

ELO Score

128K / 256K  
Context

128K 256K

Available Now

Apache 2.0 License LM Studio Ollama